

Geophysics III

Calculus Tutorial

Contents

1	Introduction	1
2	Limits	1
3	Infinitesimals	2
4	Derivatives	3
4.1	Slope	3
4.2	Anatomy of a Derivative	4
4.3	Practical Differentiation	4
4.3.1	Table of Common Derivatives	4
4.3.2	Using the Derivative Table	5
4.3.3	Practice Problems	7
4.4	Maxima and Minima	7
4.5	Multiple Derivatives	8
4.6	Partial Derivatives	8
5	Integrals	8
5.1	Areas	9
5.2	Anatomy of an Integral	9
5.3	Practical Integration	10
5.3.1	Table of Common Integrals	10
5.3.2	Using the Integral Table	10
5.3.3	Practice Problems	12
5.4	Mean of a function	13
5.5	Multiple Integrals	13

1 Introduction

Derivatives and integrals seem to frighten most students. Perhaps because most of us are taught to fear math and the further one gets in math the more it should be feared. Calculus can at times be rather complex, but in general it is relatively easy (especially derivatives) and requires learning a few simple rules and a handful of standard forms. Most derivatives and many integrals can be simplified into one of any of these standard forms, but it requires seeing patterns that allow us to transform a more complicated forms into a simple standard one. Mostly, learning how to use calculus will take a little experience but not a lot of hard work.

Calculus is the study of infinitesimals, that is the study of operations performed on functions as changes in the distance between two knowns become really really small. This allow us to determine slopes and inflection points, areas and volumes, maxima and minima, and averages. To understand calculus, we first need to understand the concept of limits and infinitesimals. We'll start there, but then continue onto derivatives and integrals which give us all the wonderful things I just mentioned.

2 Limits

Limits give us the ability to describe the class of problems where we wish to know the value of a function while approaching some known point, but cannot quite reach it. This may sound a little silly at first blush, but there are plenty of points we cannot reach, ∞ and $-\infty$ for starters. But for some functions, such as x^{-1} , we cannot evaluate the function at 0 or another some other well defined point(s). Don't believe me. Go ahead. Your calculator will cowardly refuse to answer if you try.

We write a the limit of a function at x as

$$\lim_{x \rightarrow x_0} f(x) = L, \quad (1)$$

indicates limit as x approaches x_0
limit of function at x_0^*

function of x

*where $f(x_0)$ may be undefined, but can be predicted.

So how do limits work? As an example, suppose we wanted to know what happens to $\exp(-x)$ as x approaches 1. This is a trivial case since we can easily evaluate the function to find $\exp(-1) = e^{-1}$.

Now suppose we wanted to know what happens when x becomes infinite. We cannot directly evaluate $\exp(-\infty)$ so we need a way to determine the value at ∞ , we write this problem as

$$\lim_{x \rightarrow \infty} \exp(-x) = ?, \quad (2)$$

which we would read as, the limit of $\exp(-x)$ as x approaches ∞ . If we rewrite this problem just a little,

$$\lim_{x \rightarrow \infty} \frac{1}{\exp(x)} = \frac{1}{\lim_{x \rightarrow \infty} \exp(x)}. \quad (3)$$

We can write $\exp(x)$ as an infinite series

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots. \quad (4)$$

Clearly as $x \rightarrow \infty$, this infinite series then implies $\lim_{x \rightarrow \infty} \exp(x) = \infty$ and likewise $\lim_{x \rightarrow \infty} \exp(-x) = 0$. In fact, one could prove that for any positive number greater than 1, $a^{-\infty} = 0$. The result detailed above really reiterates what we could have guessed all along since the answer is well known, but it may not always be possible to guess accurately—we'll generally stick to things that are well known in this class.

Let's return to x^{-1} as $x \rightarrow 0$. If we were to plot this function, we would see that for positive values of x , the function as small values is very large and for small negative values it is very small. This presents a problem because the two branches of the function approach different values depending on which side of 0 we approach from. Hence, for some cases, we have to indicate our direction of approach which we can do by $x \rightarrow 0^+$ from above and $x \rightarrow 0^-$ from below.

Like I mentioned before, we'll stick to cases where we know what the asymptotic value is (or easy to guess) without having to do a bunch of rigorous work. Hopefully now, you have an idea of what a limit is doing and how we write it.

3 Infinitesimals

Zeno, an ancient Greek mathematician, posed a paradox, that illustrates infinitesimals fantastically. In the course of a race, a runner will reach a point that is half the distance between the start and finish. Some time later, he is half way between the half way mark and the finish (at the quarter mark). And again he must reach a point that is half way between the quarter mark and the finish—and so on, such that he must always reach a point that is half way between a previous half way mark and the finish line without crossing. So then, how can the runner ever reach the finish line if they must always reach a point half way between the finish and a point not quite there. An infinitesimal is just that, a point that you can never quite reach. How small are they? I don't know, but I can always make a quantity smaller than a number you suggest. It is kinda like when kids discover infinity and find that some infinities are larger than others.

“Oh yeah, infinity plus one.”

“Nuh-uh, infinity times infinity.”

Calculus relies on this concept of infinitesimals, small numbers so small such that there is nothing smaller except 0 itself.

4 Derivatives

When we take a derivative we are determining how fast a function is changing with respect to something—a spatial coordinate (say z), time or whatever. You are already familiar with derivatives. Your car's speedometer reports a derivative of your position with respect to time. The slopes on a ski run are the derivative of elevation with respect to position. The cricket scoring runs per over is also a derivative with respect to overs. So derivatives are used frequently even though we may not always think of them in this way.

4.1 Slope

Derivatives are all about rates and differences. More specifically, the rate in change of a function at a point. As an example, we will look at one of the simplest functions one can draw, a line.

Recall the general equation of a line, $y = mx + b$, where m is the slope and b is the y -intercept. A derivative is very similar to our definition of slope, which for a line is defined as

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad (5)$$

where (x_1, y_1) and (x_2, y_2) are any two points on the line but $x_1 \neq x_2$. We can write these differences in x as $\Delta x = x_2 - x_1$ and similarly for y , $\Delta y = y_2 - y_1$, which gives for the slope

$$m = \frac{\Delta y}{\Delta x}. \quad (6)$$

If we reduce Δx , making it smaller and smaller, Δy will likewise shrink since the slope of the line must remain constant. If we continue to make Δx smaller such that it approaches zero, but never reaches zero, we can say that Δx is infinitesimally small. This concept of infinitesimals is fundamental to understanding both derivatives and integrals.

We express this idea of an infinitesimally small quantity as dx . Mathematically, we would say that in the limit as Δx approaches zero, the result is dx . In math speak, we write this limit as $\lim_{\Delta x \rightarrow 0} = dx$. So how small is dx ? I can't put an exact number on it. However, if you choose a value, I can always find a smaller value between your choice and zero. As a result of $dx \rightarrow 0$, Δy must also tend to zero, giving us dy . Remember for a line no matter how small dx and dy become, the ratio of the two will always yield a constant, the slope,

$$m = \frac{dy}{dx}. \quad (7)$$

We call the above result a derivative. In this case, specifically the derivative of a line. The derivative of a line is the slope of the line at every point along it. So the derivative is the rate in change of the function (y) with respect to the independent variable (x). While the derivative of a function is a constant only if the function is a line, this concept of measuring the change in a function over a very small change in the independent variable is kept.

If you are interested, look up the epsilon-delta proof online, it is the formal proof of what we have just described above.

4.2 Anatomy of a Derivative

Above, we showed that the derivative of a line is its slope. We can extend this idea of the slope at a point for any function, $f(x)$, more generally. To express this idea, we write

$$\frac{d}{dx} f(x) = f'(x). \quad (8)$$

where $f(x)$ is an arbitrary function of x , x is the independent variable, and $f'(x)$ is the derivative of $f(x)$ and also a function of x . The $\frac{d}{dx}$ is called the differential operator. If you are unfamiliar with the term operator, an operator is a symbol or collection of symbols when taken as a whole that perform a defined mathematical task. For example, the ' $\frac{d}{dx}$ ' operator indicates that we take the derivative of a function with respect to x just as a '+' means that we add the quantities on either side.

4.3 Practical Differentiation

We could of course spend the time—as one might in a math course—by deriving and proving the rules and standard forms for differentiation. However, since mathematicians have already done the hard work for us, we can just focus on using them instead. All or at least nearly all of the derivatives and rules we will need for this course can be found in the math appendices under the table of derivatives.

When I say that all the derivatives that we will need for the course are found within the table of derivatives, it does not necessarily mean that they are symbol for symbol the same. Every equation you will encounter can be rearranged into one of these common forms. I've included the table again below for convenience, but how do we use them?

4.3.1 Table of Common Derivatives

Derivatives of a few common functions (note $u = u(x)$, $v = v(x)$, a and n are constants):

$$\frac{d}{dx}(a) = 0$$

$$\frac{d}{dx}(au) = a \frac{du}{dx}$$

$$\frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$\frac{d}{dx}u(v) = \frac{du}{dv} \frac{dv}{dx}$$

$$\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$$

$$\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = v^{-2} \left(v \frac{du}{dx} - u \frac{dv}{dx} \right)$$

$$\frac{dy}{dx} = \left(\frac{dx}{dy} \right)^{-1}$$

$$\frac{d}{dx}\left(\frac{1}{u^n}\right) = -\frac{n}{u^{n+1}} \frac{du}{dx}$$

$$\frac{d}{dx}(\log u) = \frac{1}{u} \frac{du}{dx}, \quad u > 0$$

$$\frac{d}{dx}(\cos u) = -\sin u \frac{du}{dx}$$

4.3.2 Using the Derivative Table

The derivative table above can be used to take the derivatives of many more functional forms than those shown in the table. In fact, the table is much more general than it appears. The letters ' a ' and ' n ' in the table indicate constants. These constants can be anything—a number, a collection of constants, a function of an independent variable—so long as it isn't a function of the variable of differentiation. The letters ' u ' and ' v ' are both functions of x , which can be any arbitrary function so long as it is differentiable (we discussed this on problem set 1). But this table doesn't just work for derivatives with respect to x . They can be with respect to any independent variable.

So what to do if our problem does not look like one of the derivatives listed on the table?

1. Find a set of functions and constants which transform the original function to look like a form in the table; that is, by choosing an appropriate function, $u(x)$ and/or $v(x)$, and/or constants, a and/or n .
2. Copy the derivative from the table that matches the transformed equation.
3. Substitute our original constants and functions into the standard derivative solution.
4. Perform the next derivative, repeating the previous three steps until $\frac{du}{dx} = \frac{dx}{dx} = 1$ or $\frac{da}{dx} = 0$.

One thing you will discover is that for a large majority of these problems, we will have to take several derivatives.

Example 4.1. Suppose,

$$f(x) = \frac{1}{(x^2 + c^2)^{1/2}},$$

where c is a constant. Find

$$\frac{d}{dx} \left(\frac{1}{\underbrace{(x^2 + c^2)}_{= u(x)} \underbrace{^{1/2}}_{= n}} \right).$$

If you look at the table of derivatives, you will not find this exact solution listed. However, looking at the table and our function we can see that if we write our function as

$$\begin{aligned} f(x) &= \frac{1}{u^n}, \\ u &= x^2 + c^2, \\ n &= \frac{1}{2}. \end{aligned}$$

The derivative of this function is in our table of derivatives,

$$\frac{d}{dx} \left(\frac{1}{u^n} \right) = -n \frac{1}{u^{n+1}} \frac{du}{dx}.$$

Substituting the original transformed values back in our transform, we find

$$\frac{d}{dx} \left(\frac{1}{u^n} \right) = -\frac{1}{2} \frac{1}{(x^2 + c^2)^{1/2+1}} \frac{d}{dx} (x^2 + c^2).$$

So you'll see we have another derivative that we need to determine, but at least we have simplified our original derivative. Using the same procedure again, we need to transform the derivative term into one in the table.

$$\begin{aligned} \frac{d}{dx} \frac{1}{(x^2 + c^2)^{1/2}} &= -\frac{1}{2} \frac{1}{(x^2 + c^2)^{3/2}} \frac{d}{dx} (\underbrace{x^2}_{= u(x)} + \underbrace{c^2}_{= v(x)}). \end{aligned}$$

Looking at the table of derivatives, we see that we can write our derivative in the form

$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}.$$

Therefore,

$$\frac{d}{dx} \frac{1}{(x^2 + c^2)^{1/2}} = -\frac{1}{2} \frac{1}{(x^2 + c^2)^{3/2}} \left(\frac{d}{dx} \overset{= u(x)}{\overset{2}{x}} + \frac{d}{dx} \overset{= a}{c^2} \right).$$

Now we have two derivatives that we need to evaluate. For the first, we use

$$\begin{aligned} \frac{d}{dx} u^n &= n u^{n-1} \frac{du}{dx} \\ &= 2x^{2-1} \frac{dx}{dx}. \end{aligned}$$

Since we can see from the table that $\frac{dx}{dx} = 1$, the result of the derivative $\frac{dx^2}{dx} = 2x$. Turning our attention to the second derivative, we see from the table of derivatives that $\frac{da}{dx} = 0$ when a is independent of x . Now we've completed all the derivatives we need to perform and we can focus on simplifying our expression,

$$\begin{aligned} \frac{d}{dx} \frac{1}{(x^2 + c^2)^{1/2}} &= -\frac{1}{2} \frac{1}{(x^2 + c^2)^{3/2}} (2x + 0), \\ \frac{d}{dx} \frac{1}{(x^2 + c^2)^{1/2}} &= -\frac{x}{(x^2 + c^2)^{3/2}}, \end{aligned}$$

and we are done.

This process may have seemed long and laborious, but the more you do them, the faster you will become. Also you'll start to see several steps ahead, which will allow you to more rapidly discover the solution.

We could go one set further and ask what the slope of a function is at a particular point, say $x = x_0$. In this case, we determine the derivative and then evaluate our result with x replaced by the value x_0 .

4.3.3 Practice Problems

I will give you some time to try these and then we'll do them together as part of the tutorial.

Problem 4.1. Plate Cooling

To find the heat flow from the temperature, we need to take the derivative of the temperature with respect to depth. For plate cooling of the lithosphere, we need the derivative of the term

$$\frac{d}{dz} \frac{1}{n} \sin\left(\frac{n\pi z}{L}\right)$$

Problem 4.2. Seismic Reflection

To find the minimum travel-time due to a seismic reflection off a boundary, we need to take the

derivative of $t(x)$ as below,

$$\frac{d}{dx} v_1^{-1} [(h^2 + x^2)^{1/2} + (h^2 + (d - x)^2)^{1/2}]$$

Problem 4.3. Decompression Melting

To model partial melting, you need to take the derivative of the melt depression term,

$$\frac{d}{dF} K \left(\frac{X_0}{D + F(1 - D)} \right)^{\gamma}.$$

Problem 4.4. Gravity on an Ellipsoid

As part of the derivation to determine the gravity of an ellipsoid (remember we make an ellipsoidal correction to gravity), we need to differentiate the equation of an ellipsoid,

$$\frac{d}{d\theta} \frac{1 - f}{1 - f \sin^2 \theta}. \quad (9)$$

4.4 Maxima and Minima

One of the really useful applications of a derivative is its use to determine the locations of maxima and minima of a function. While some functions extend to $\pm\infty$, there may be individual points where a function decreases in value before increasing (minima) or increases in value before decreasing (maxima). For a maxima, logic tells us that there must be some point along the path for which the slope on the left is positive and the slope on the right is negative. If so, there must be a point where the slope changes sign and, therefore, must be zero. So if we want to know where this point occurs, then we simply find our expression for the derivative and set it equal to zero (i.e., $\frac{df(x)}{dx} = 0$) and then solve for x . The same must also be true for minima. So we may not know, simply by taking the derivative, whether the point we found is a maximum or minimum, but we can easily check by plotting the function.

An extension of the maxima and minima is finding the points for which the slope of a function is maximum or minimum (these points are called inflection points). How might we find these points using derivatives?

4.5 Multiple Derivatives

As alluded to previously, it is possible to take the derivative of a function multiple times. The second derivative of a function is quite commonly used to describe physical systems. We often call the second derivative the curvature because it tells us how fast the slope is changing.

To take multiple derivatives, all you need to do is follow the rules to find the derivative of a function, then take the derivative of the result, and so on until the desired derivative is reached. To indicate higher order derivatives, we include the order of the derivative, e.g., for a second order derivative,

$$\frac{d}{dx} \frac{df(x)}{dx} = \frac{d^2 f(x)}{dx^2}. \quad (10)$$

I cannot think of any derivatives higher than order four in geoscience, and those are uncommon, used only in flexural problems. Generally, you won't need to take more than one or two.

4.6 Partial Derivatives

Suppose we had a function that had multiple independent variables, i.e., $f(x, y, z)$. For example, gravity fields due to a sphere, elevation, and temperatures within the Earth. We can also take the derivatives of these functions. However, when we do so, we specify the independent variable or multiple independent variables to differentiate with respect to. We call these derivatives partial derivatives. When we take partial derivatives, we change our notation slightly, exchanging our ‘ d ’ with a ‘ ∂ ’. Because each variable is independent of the others, when we take a derivative with respect

5 Integrals

Integrals are the opposite of a derivative, and are called by some an anti-derivatives. The fundamental theorem of calculus defines an integral this way, i.e.,

$$f(x) = \frac{d}{dx} \int_0^x f(x') dx',$$

which basically says the integral is the opposite of a derivative.

We use integrals to determine the area or volume defined by some boundary. A submarine uses its velocity and the time it travels in a given direction to calculate the position of a ship. The electrical energy you consume is determined by the integral of your power consumption, which is used to figure your bill from the power company. The color of an object is results from the integral of the spectrum of light reflected off the object and the sensitivity of the cones in your eyes.

5.1 Areas

To perform a derivative, we determine the ratio between small differences in a function with small differences in the independent variable. An integral does the opposite, determining the sum of small areas. The areas are the products between small differences of the independent variable and the average of the function over the same intervals, i.e.,

$$F(x) = \sum_{i=1}^N \frac{(f(x_i) + f(x_{i-1}))/2}{1} (x_i - x_{i-1}).$$

as the interval $\Delta x_i = x_i - x_{i-1}$ gets smaller and smaller, it becomes an infinitesimal just as we described previously in the derivative section. If we chose our x_i ’s such that they were all the same size, then over the interval from a to b , we can write

$$\Delta x_i = \Delta x = \frac{b - a}{N}.$$

As the interval becomes smaller and smaller, i.e., $N \rightarrow \infty$, $f(x_i) \approx f(x_{i-1})$. This allows us to write

$$F = \sum_{i=1}^N f(x_i) \Delta x_i = \int_a^b f(x) dx.$$

In fact, the integral symbol, $\int$, was originally an S to signify that an integral was a sum, so the symbol itself is a stylized S retaining that connection.

5.2 Anatomy of an Integral

Above, we showed that the integral of a line is related to its area beneath a curve. We can extend this idea of the area over any interval $[a, b]$ for a function, $f(x)$, more generally. To express this idea, we write the definite integral

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a) \quad (11)$$

integral with limits $[a, b]$
integral of $f(x)$

variable of integration, x

integrand, function of x
evaluated from a to b

where $f(x)$ is an arbitrary function of x , x is the independent variable, and $F(x)$ is the integral and also a function of x . An integral is often taken over a bounded region from $[a, b]$. Not all integrals have limits. In these cases, the integral is called an indefinite integral and the solution will include an arbitrary constant.

5.3 Practical Integration

Just as there is for derivatives, a table of integrals can be utilized to simplify complex integrals. The steps to using them and find a solution are similar to those we used for a derivative. Although, there is an extra step necessary to transform the infinitesimals from one into another.

5.3.1 Table of Common Integrals

Integrals of a few common functions (note $u = u(x)$, $v = v(x)$, a and n are constants and C is an arbitrary constant due to integration):

$$\int a dx = ax + C$$

$$\int au dx = a \int u dx$$

$$\int u dv = uv - \int v du$$

$$\int u^n du = \frac{1}{n+1} u^{n+1} + C, n \neq -1$$

$$\int e^u du = e^u + C$$

$$\int \sin u du = -\cos u + C$$

$$\int x dx = \frac{1}{2} x^2 + C$$

$$\int (u + v) dx = \int u dx + \int v dx$$

$$\int u(v) dx = \int u(v) \left(\frac{dv}{dx} \right)^{-1} dv$$

$$\int \frac{1}{u} du = \log u + C$$

$$\int \log u du = u \log u - u$$

$$\int \cos u du = \sin u + C$$

5.3.2 Using the Integral Table

We use the integral table in a similar way we use the derivative table. If our form is in the table, we copy it and we're done. But if not, there are multiple methods for determining the integral to a function. We will discuss a few here. This first integration method is by substitution using a transform function similar to our method for derivatives above,

1. Find an equation $u(x)$ which simplifies our function.
2. Determine $\frac{du}{dx}$ and solve for du to replace the variable of integration.
3. Copy the integral from the table that matches our transformed function.
4. Substitute our constants and functions into the standard integral solution.
5. If there are any integrals remaining, repeat steps above or use another method.

Example 5.1. Integrate

$$\int x e^{-x^2} dx.$$

We need a function that can transform our problem above into one that is located in the table. To do so, we look for the integral that has an exp term. It may take a little practice to see which function to use as a transform, but in this case, we use $u = -x^2$.

We also need to change our variable of integration to u . To do so, we need to start with the derivative of our transform with respect to x , i.e.,

$$\begin{aligned}\frac{du}{dx} &= -2x \\ du &= -2x dx \\ -\frac{du}{2x} &= dx.\end{aligned}$$

Now we can transform the integral in to a simpler form,

$$\begin{aligned}\int x e^{-x^2} dx &= \int x e^u \left(-\frac{du}{2x} \right) \\ &= \int -\frac{1}{2} e^u du.\end{aligned}$$

Solving this integral and substituting back in our original function, we find

$$\begin{aligned}-\frac{1}{2} \int e^u du &= -\frac{1}{2} e^u \\ &= -\frac{1}{2} e^{-x^2}.\end{aligned}$$

The second integration method requires integration by parts. To integrate by parts,

1. Find two functions, $u(x)$ and $v(x)$ so that we can write our expression as $\int u dv$. We'll need $\frac{dv}{dx}$ in order to make this substitution.
2. Write the solution to the integration by parts form found in the table, i.e., $\int u dv = uv - \int v du$.
3. Solve the integral $\int v du$ by the first or third method.

Example 5.2. Integrate

$$\int x \cos x dx.$$

Choosing $u = x$ and $v = \sin x$, we can determine dv ,

$$\begin{aligned}\frac{dv}{dx} &= \cos x \\ dv &= \cos x \, dx\end{aligned}$$

Using the integral solution for parts,

$$\int u \, dv = uv - \int v \, du, \quad (12)$$

substitute in our equations,

$$\int x \cos x \, dx = x \sin x - \int \sin x \, dx.$$

Next, we solve the new integral using the form from the integral table and we are done,

$$\int x \cos x \, dx = x \sin x + \cos x + C,$$

where C is an arbitrary constant as a resulting from the indefinite integral.

The third integration method is similar to the substitution method, but specifically applies to trigonometric substitution. This method is essentially a coordinate transformation. To do this, we apply the following substitutions

$$a^2 - x^2 = a^2 \sin^2 \theta \quad (13)$$

$$x^2 + a^2 = a^2 \sec^2 \theta \quad (14)$$

$$x^2 - a^2 = a^2 \frac{1}{\cos^2 \theta}. \quad (15)$$

Once we make our substitution, we follow one of the other two integration methods above.

5.3.3 Practice Problems

Problem 5.1. Steady-State Temperature

To find the temperature in a half-space, we need to integrate the thermal gradient with depth,

$$\int \left(\frac{q_0}{k} - \frac{A}{k} z \right) dz,$$

where q_0 , A , and k are constants.

Problem 5.2. Radioactive Decay

To find the number of parent radionuclides present due to radioactive decay, we need to integrate both sides of the following equation,

$$\int_{N_0}^N \frac{dN'}{N'} = - \int_0^t \lambda \, dt',$$

and once complete, solve for N .

Problem 5.3. Buried Slab

This integral is similar to one we needed for our buried slab problem,

$$\int_0^R \frac{r}{(r^2 + z^2)^{3/2}} dr.$$

Problem 5.4. Mass Flux

The mass flux of asthenosphere (mass passing a given point per unit time) moved by a tectonic plate can be expressed as

$$\int_0^h \rho u_0 \frac{y}{h} dy.$$

Problem 5.5. Entropy

In petrology, the entropy is a key thermodynamic parameter. The change in entropy is related to temperature as

$$\Delta S = \int_{T_1}^{T_2} \frac{C_P}{T} dT. \quad (16)$$

5.4 Mean of a function

In addition to finding areas and volumes, we can use integrals to find the mean of a function. For example, take a function between two points, say x_1 and x_2 . Assuming our function is continuous, logic will tell us that our function must pass through the mean value at least once on our chosen interval. This is called the mean value theorem. To find the mean value of our function is sort of like taking the arithmetic mean of a discrete set of point and dividing by the number of points only we take the sum over infinitely small intervals and divide by the total width of the intervals. To write this as an integral,

$$\overline{f(x)} = \frac{1}{(x_2 - x_1)} \int_{x_1}^{x_2} f(x) dx, \quad (17)$$

where the bar over the function indicates the mean of the function.

5.5 Multiple Integrals

Just as we had for derivatives, we can take multiple integrals. We might do this for the accelerations that we measure from an earthquake to determine the change in displacement of the ground surface during an earthquake.

To determine a multiple integral, we would write

$$\int \left(\int f(x) dx \right) dx = \iint f(x) dx^2.$$

If we are taking multiple derivatives, but not for the same coordinate, then we are finding the integral of a surface or a volume, i.e.,

$$\begin{aligned} \text{surface: } & \iint f(x, y) dx dy \\ \text{volume: } & \iiint f(x, y, z) dx dy dz \end{aligned}$$

Often when we evaluate surface or volume integrals it is advantageous to make a transform to another coordinate system first in order to exploit symmetries simplify the integration. For example,

it is easier to integrate the gravity field due to a sphere in spherical coordinates than in Cartesian coordinates.